Outlook

Re: Using birds to gather plastic waste

Fra Jonathan Wright <jonathan.wright@ntnu.no>
Dato ma. 09.02.2026 12:39
Til Simon Andreas Degelmann <simonade@stud.ntnu.no>

Hei Simon et al.,

I would agree that crows are you best bet here, as I said previously, but you don't need too many operant stimuli, like lights and sounds. It is essentially the food reward that drives the behaviour. You could add a simple sound or light cue to confirm an appropriate delivery prior to the reward delivery, but it probably won't make too much difference beyond the initial training phase. Indeed, it is the training phases that are the real problem, because you initially need to attract the birds with bait food and then take them through a series of steps to complete their learning of the desired activity. This training is often difficult and time consuming, because it has to be done manually or with a sophisticated flexible automatic feeder set-up, and this is something only a few people already working with animals like this probably have the skills to do with any efficiency.

Hence, I don't think that this represents a long-term or large-scale solution, because all the effort involved might as well instead be used picking up the litter yourselves! Even if you can get a few birds doing the right behaviours, you will have to train every new set of birds, since this behaviour won't be culturally transmitted within flocks. There is also the obvious problem of the birds getting bored and stopping doing the desired behaviour after a while, even for your best performing individual and if you keep introducing novel food types. This is because they have interesting and useful social lives, additional diverse foraging activities and breeding to think about, which they have evolved to value more than a single food source, however reliable. So, I don't think this will work in the long term or at any useful scale.

If you need evidence that what I say is true then just search for these type of schemes on the internet, mostly for collecting cigarette butts - there's a nice summary in 2022 here: https://corvidresearch.blog/2022/02/05/butts-for-nuts-can-crows-do-our-dirty-work-and-should-they/. They mention the 2022 pilot project in Sweden, which like all the previous projects of this type didn't succeed and wasn't taken up by the local authorities (despite a resurgence of even more internet misinformation on this in 2025). As I say, these things can work for a short time on a small number of birds that have been intensively trained, but they don't represent a long-term, wide-scale or sustainable solution to the problem of litter. The technical aspects of the operant machinery, which you guys have an interest in, is the easy part to build and get functioning. It's the natural behaviour of the birds that is your problem.

Sorry, again, for being so negative. Let me know if you need any further input on this topic.

Mvh,

Jon

On 9 Feb 2026, at 11:15, Simon Andreas Degelmann <simonade@stud.ntnu.no> wrote:

Hello again Jon.

Thank you for your previous answer! Your Email was very insightful, and we have shifted our design towards crows, as you suggested. We would love to learn more about this species of bird, and how we may understand and use their psychology and behaviors to our advantage.

Humans nowadays are very susceptible to social media and gambling, this being in part because of the user interfaces used to interact with the medium (bright lights, quicky sounds and instant feedback). We were hoping to apply the same principles to make crows more interested in our product, and we were curious about the following:

Would it be beneficial to expose the crows interacting with our device to sounds, light and other sensory stimulation to increase their feeling of reward? This of course being in addition to the food they are getting.

Is there a specific type of food they prefer? If this item is unhealthy in the long term, it could only be rewarded a smaller fraction of the time, adding anticipation in the crows when using the device.

Any insight into the matter is highly appreciated!

Regards E. Born, S. Degelmann, E. Alsos, M. Eriksson, M. Helgeland, E. Nestaas.

Fra: Jonathan Wright <jonathan.wright@ntnu.no>
Sendt: mandag 26. januar 2026 18:24
Til: Simon Andreas Degelmann <simonade@stud.ntnu.no>
Emne: Re: Using birds to gather plastic waste

Hei Simon,

I think you've already received some useful comments from Thor Harald and Sigurd on this innovative idea, and I largely agree with them. I've been involved for many years in studies of avian foraging behaviour and experimental training/taming of various wild bird species, and although this is a nice idea, I'm not sure how practical it is.

A major issue is the particular species of bird involved, because there are large differences in training, learning and foraging between seabirds, like gulls/waders, and the various species of passerines, like magpies/crows/ravens, that have been successfully used in various experimental studies of this type. Unfortunately, if you want pieces of plastic to be recovered from a range of marine habitats beyond just certain areas of coastline, then a species like seagulls might be the only way to do it.

There is a lot of work needed to train birds to recover non-food items and return them to some central place and exchange them for food items, because this is very different from any of their natural behaviours. Perhaps it can be done with one or two individuals, probably crows, but it might be too hard to train lots of individuals across different locations, and they are very unlikely to spontaneously learn it from each other via cultural transmission within their foraging flocks.

So, I'm sorry to be a bit negative here, but the easiest part of this would be the technical aspects of building the automated machine to do the exchanging. It is the natural history of birds that makes it unlikely to work without prohibitively large amounts of work. Let me know if you have any further questions.

Mvh,

Jon

> On 26 Jan 2026, at 11:05, Simon Andreas Degelmann <simonade@stud.ntnu.no> wrote:

We are a group of students at NTNU, from the study program Electronic System Design and Innovation. This semester we are working on an innovation-project concerning plastic in the ocean. At this time, we want to explore the possibility of allowing animals (mainly birds) to contribute. We are working on a product that allows birds to deliver plastic they find in nature, in exchange for food. The product will use machine learning to only reward birds that deliver unnatural items such as plastic or other trash. Is this possible to achieve? How does one proceed with such a process, from an ecological perspective? Could this be harmful, both in the short and long term? We've planned to "use" seagulls for this project, is this a good choice? Are there better alternatives? Can such birds be trained to do such tasks and if so, how?

Hope to hear from youNederst I skjemaet

Regards E. Born, S. Degelmann, E. Alsos, M. Eriksson, M. Helgeland, E. Nestaas.

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